

OITA, Japan

WMO: 478150

Lat: 33.2362N

Lon: 131.6200E

Elev: 44

StdP: 14.67

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 31.5	(c) 33.3	(d) 16.3	(e) 12.6	(f) 38.7	(g) 18.5	(h) 14.0	(i) 39.9	(j) 17.0	(k) 44.9	(l) 15.9	(m) 44.3	(n) 6.6	(o) 180	(p) 0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.9	(c) 92.5	(d) 77.8	(e) 90.8	(f) 77.5	(g) 88.8	(h) 77.0	(i) 79.8	(j) 88.2	(k) 78.9	(l) 87.1	(m) 78.0	(n) 85.8	(o) 8.5	(p) 10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.5	143.1	83.9	76.6	138.7	83.2	75.7	134.8	82.6	43.4	88.5	42.4	87.2	41.5	86.1	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 15.8	(b) 13.8	(c) 12.1	(e) 28.1	(f) 96.1	(g) 1.9	(h) 1.6	(i) 26.7	(j) 97.2	(k) 25.6	(l) 98.1	(m) 24.6	(n) 99.0	(o) 23.2	(p) 100.2		
(a) 15.8	(b) 13.8	(c) 12.1	(e) 24.9	(f) 81.1	(g) 2.1	(h) 1.2	(i) 23.4	(j) 81.9	(k) 22.1	(l) 82.6	(m) 20.9	(n) 83.3	(o) 19.4	(p) 84.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.6	43.6	45.0	50.3	58.8	67.1	73.1	81.0	82.7	76.2	66.9	56.9	47.8	
	DBStd	14.20	4.22	5.03	5.24	5.13	3.93	4.17	3.84	3.14	4.47	4.77	5.38	5.02	
	HDD50	533	204	156	60	3	0	0	0	0	0	0	6	104	
	HDD65	2711	664	561	456	192	23	0	0	0	0	33	248	534	
	CDD50	5115	5	16	70	268	531	692	962	1015	786	523	212	35	
	CDD65	1817	0	0	0	7	89	243	497	550	336	91	4	0	
	CDH74	15461	0	0	3	29	329	1247	5006	6121	2424	298	4	0	
	CDH80	5205	0	0	0	2	26	256	1852	2441	603	25	0	0	
Wind	WSAvg	5.8	6.5	6.3	6.2	5.8	5.6	4.8	5.2	5.6	5.5	5.7	5.9	6.5	
Precipitation	PrecAvg	66.5	1.9	2.5	3.9	5.0	5.7	11.3	9.5	6.6	9.5	5.6	2.8	1.5	
	PrecMax	112.7	7.2	6.9	12.7	9.7	14.1	24.8	27.6	28.8	27.6	24.4	8.2	5.8	
	PrecMin	42.3	0.0	0.3	1.0	0.7	0.8	1.7	1.5	0.1	0.2	0.6	0.2	0.0	
	PrecStd	13.9	1.6	1.6	2.0	1.9	3.0	5.3	6.0	5.6	6.5	5.2	2.0	1.3	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.2	66.0	70.6	77.4	83.1	89.8	94.3	95.4	91.3	83.1	73.5	64.8	
		MCWB	51.7	58.0	57.8	62.9	65.8	76.0	77.7	78.5	76.8	72.4	63.7	56.2	
	2%	DB	56.5	60.0	65.9	73.7	79.9	85.4	92.2	93.2	88.3	79.5	70.6	61.5	
		MCWB	49.0	52.5	55.9	60.2	64.6	73.4	77.7	78.1	75.7	69.5	62.3	53.8	
	5%	DB	53.9	56.6	62.6	70.6	77.7	82.9	90.4	91.5	85.7	77.2	68.3	59.1	
		MCWB	46.9	49.6	53.3	59.0	63.9	71.9	77.4	77.8	74.5	67.8	60.8	51.8	
	10%	DB	51.8	53.9	59.8	68.0	75.3	80.3	88.3	89.6	83.4	75.1	66.2	56.9	
		MCWB	45.2	47.0	52.0	57.7	63.3	70.3	77.0	77.4	73.6	66.4	59.2	50.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	59.3	61.2	67.5	72.0	77.3	80.8	81.3	79.1	75.3	67.5	59.6
			MCDB	58.6	65.3	66.2	70.7	76.8	85.5	89.2	90.6	86.2	79.9	71.1	62.0
2%		WB	51.1	54.7	58.3	64.4	69.5	75.7	79.6	80.1	77.5	73.0	64.9	56.0	
		MCDB	54.1	58.8	63.2	69.4	73.9	82.6	87.9	88.9	84.0	77.5	68.3	59.4	
5%		WB	48.2	51.3	55.5	62.2	67.6	74.5	78.6	79.2	76.5	70.4	62.8	53.4	
		MCDB	52.6	54.9	60.7	68.0	72.9	80.4	86.9	87.6	82.6	75.0	66.5	57.5	
10%		WB	45.9	48.5	53.0	60.1	66.1	73.1	77.7	78.4	75.5	68.1	60.5	51.0	
		MCDB	50.9	52.8	58.3	65.8	72.0	78.3	85.6	86.4	81.5	72.9	65.1	55.9	
Mean Daily Temperature Range		5% DB	MDBR	13.4	13.7	14.6	15.4	15.0	11.6	12.0	12.9	11.9	13.2	14.0	13.8
			MCDBR	17.0	17.7	19.1	19.8	18.9	15.3	15.1	15.5	14.4	14.9	15.7	16.7
	MCWBR		11.6	12.5	12.2	11.0	9.3	6.4	5.5	5.5	6.0	8.0	10.1	11.2	
	5% WB	MCDBR	14.6	15.3	16.1	15.0	13.7	11.8	13.5	14.1	12.3	11.0	12.2	13.7	
		MCWBR	11.8	12.5	12.3	10.0	8.6	5.6	5.4	5.6	5.7	7.1	9.0	11.0	
Clear-Sky Solar Irradiance	taub	0.419	0.462	0.525	0.549	0.557	0.567	0.565	0.538	0.508	0.464	0.428	0.410		
	taud	2.101	2.018	1.868	1.865	1.910	1.966	2.014	2.082	2.125	2.161	2.175	2.145		
	Ebn at Noon	242	245	239	240	240	237	236	241	242	242	239	237		
	Edh at Noon	41	49	61	64	62	58	56	51	47	42	37	37		
All-Sky Solar Radiation	RadAvg	770	972	1299	1530	1628	1354	1558	1612	1296	1088	836	698		
	RadStd	66	86	84	150	184	182	214	158	145	117	80	47		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92
(47) Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page